Module 7 — Nested-loop and pattern-printing questions

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Questions

Question 1 =

Question 1 — Print a Solid Square

Difficulty: Very Easy

Problem statement: Given a positive integer n, print a square containing n rows and n columns of asterisks (*). Do not print spaces between asterisks.

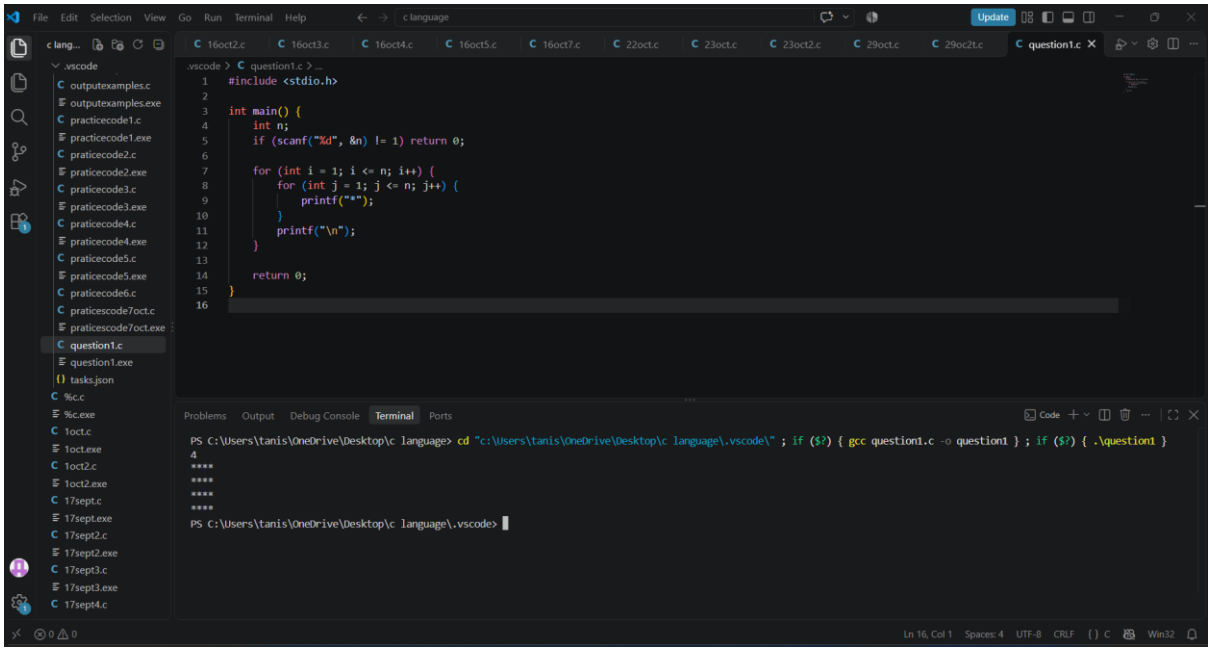
Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, each containing exactly n asterisks.

Sample Input	Sample Output
4	**** **** **** ****

Hint: Use one loop for rows and one loop for columns.

ANSWER 1 =


File Edit Selection View Go Run Terminal Help
c language
Update
c lang...
.vscode
outputexamples.c
outputexamples.exe
practicecode1.c
practicecode1.exe
praticecode2.c
praticecode2.exe
praticecode3.c
praticecode3.exe
praticecode4.c
praticecode4.exe
praticecode5.c
praticecode5.exe
praticecode6.c
praticescode7oct.c
praticescode7oct.exe
question1.c
question1.exe
tasks.json
%c.c
%c.exe
1oct.c
1oct.exe
1oct2.c
1oct2.exe
17sept.c
17sept.exe
17sept2.c
17sept2.exe
17sept3.c
17sept3.exe
17sept4.c
16oct2.c
16oct3.c
16oct4.c
16oct5.c
16oct7.c
22oct.c
23oct.c
23oct2.c
29oct.c
29oc2t.c
question1.c
.vscode > question1.c > ...
#include <stdio.h>
int main() {
int n;
if (scanf("%d", &n) != 1) return 0;
for (int i = 1; i <= n; i++) {
for (int j = 1; j <= n; j++) {
printf("*");
}
printf("\n");
}
return 0;
}
Problems Output Debug Console Terminal Ports
Code
PS C:\Users\tanis\OneDrive\Desktop\c language> cd "c:\Users\tanis\OneDrive\Desktop\c language\.vscode\" ; if ($?) { gcc question1.c -o question1 } ; if ($?) { .\question1 }
4
****
****
****
****
PS C:\Users\tanis\OneDrive\Desktop\c language\.vscode>
Ln 16, Col 1
Spaces: 4
UTF-8
CRLF
C
Win32

Question 2 =

Question 2 — Print a Right-Angled Triangle

Difficulty: Very Easy

Problem statement: Given n, print a left-aligned triangle. Row i must contain exactly i asterisks, for i from 1 to n.

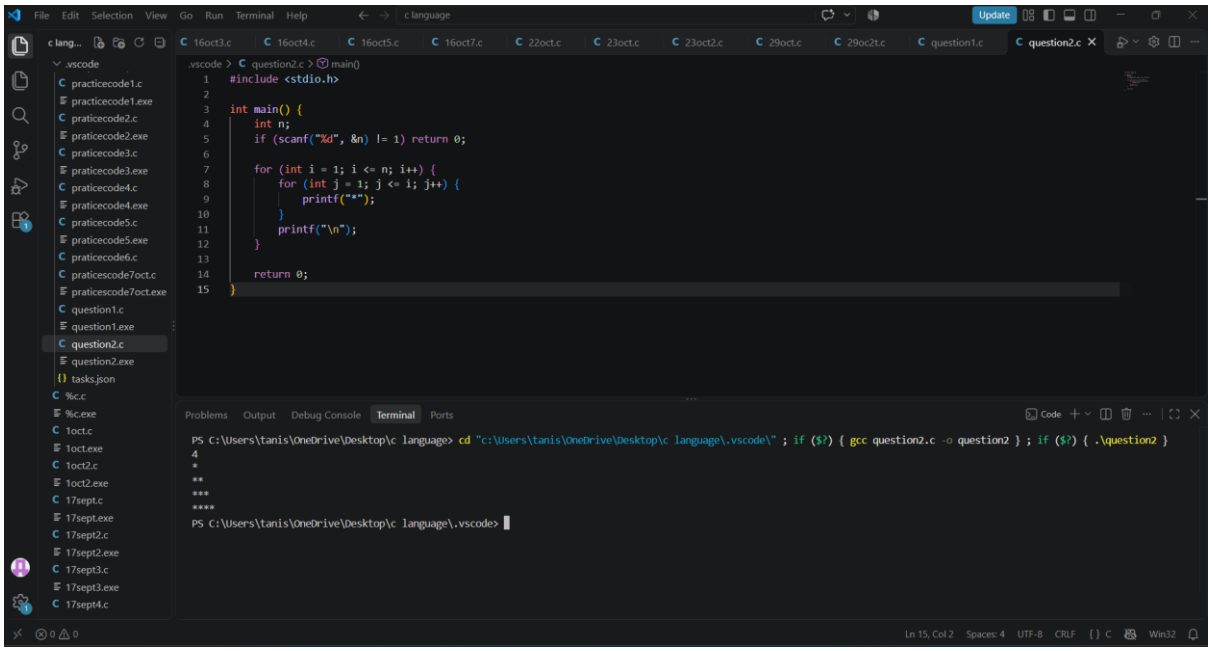
Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines. Line i contains i asterisks.

Sample Input	Sample Output
4	* ** *** ****

Hint: The inner loop limit depends on the current row.

ANSWER



Question 3 =

Question 3 — Print an Inverted Triangle

Difficulty: Easy

Problem statement: Given n, print a left-aligned inverted triangle. The first row contains n asterisks; each following row contains one fewer.

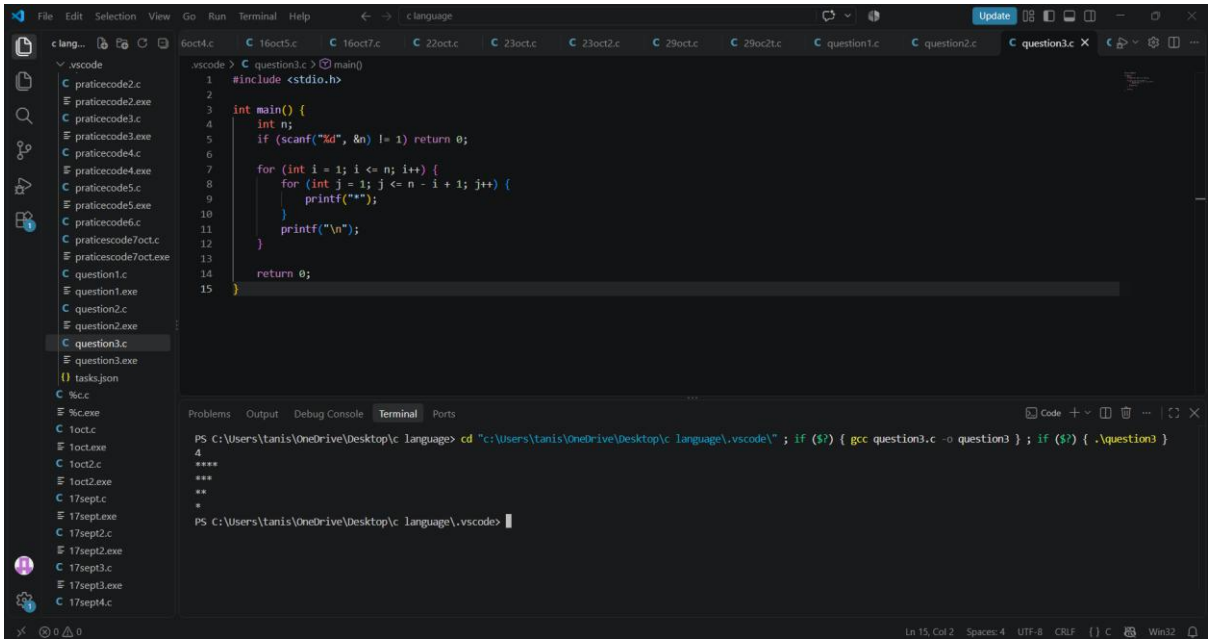
Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, with the number of asterisks decreasing from n to 1.

Sample Input	Sample Output
4	**** *** ** *

Hint: For row i, print n - i + 1 asterisks.

ANSWER 3 =



Question 4 =

Question 4 — Print a Number Triangle

Difficulty: Easy

Problem statement: Given n, print n rows. On row i, print integers from 1 through i, separated by one space.

Input format: One integer n (1 ≤ n ≤ 20).

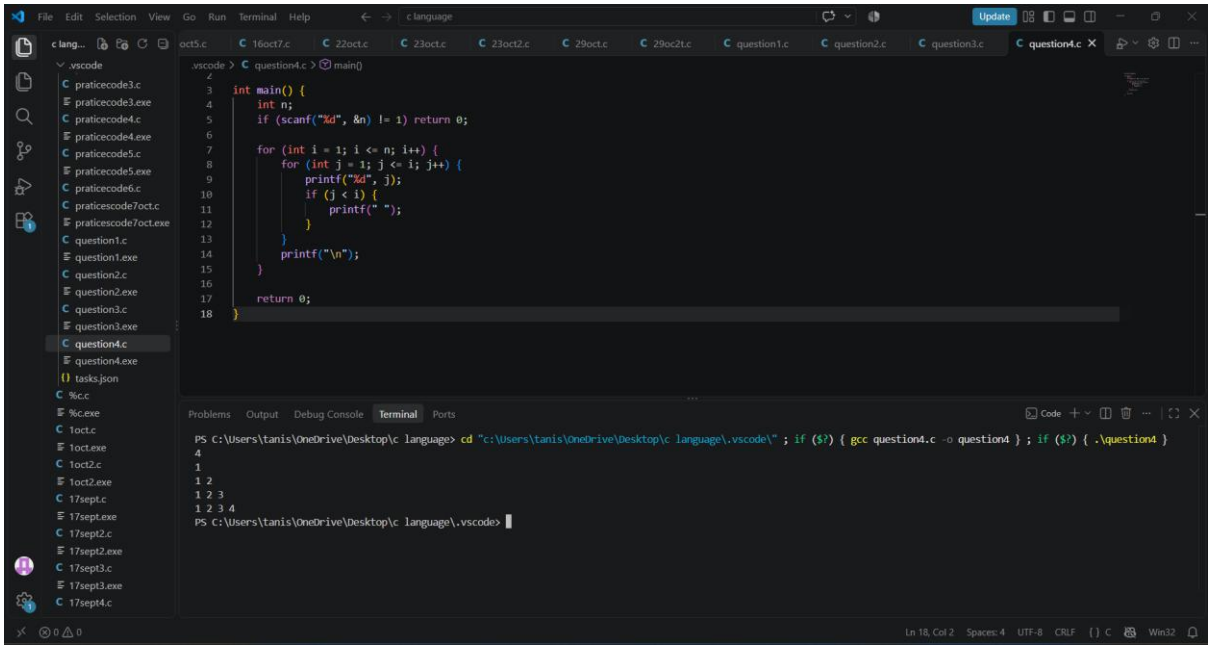
Output format: Print n lines with numbers separated by one space and no trailing spaces.

C Programming Practice Coding Platform Question Bank Set 1

Sample Input	Sample Output
4	1 1 2 1 2 3 1 2 3 4

Hint: The inner loop prints from 1 through the current row number.

ANSWER



Question 5 =

Question 5 — Print a Repeated-Number Triangle

Difficulty: Easy

Problem statement: Given n, print n rows. On row i, print the number i exactly i times, separated by one space.

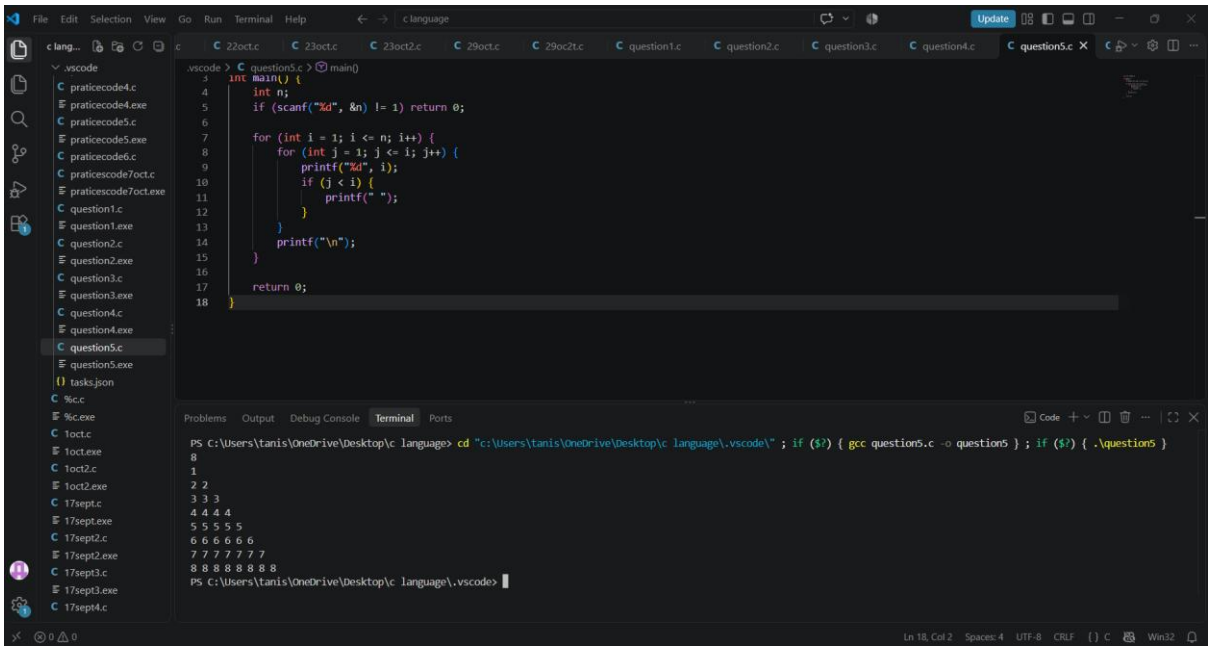
Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines; row i contains i copies of i.

Sample Input	Sample Output
4	1 2 2 3 3 3 4 4 4 4

Hint: Print the row counter during each inner-loop iteration.

ANSWER



Question 6 =

Question 6 — Print a Consecutive-Number Triangle

Difficulty: Easy

Problem statement: Print consecutive positive integers across triangular rows. Row 1 has one number, row 2 has two, and so on. Counting starts at 1 and continues across rows.

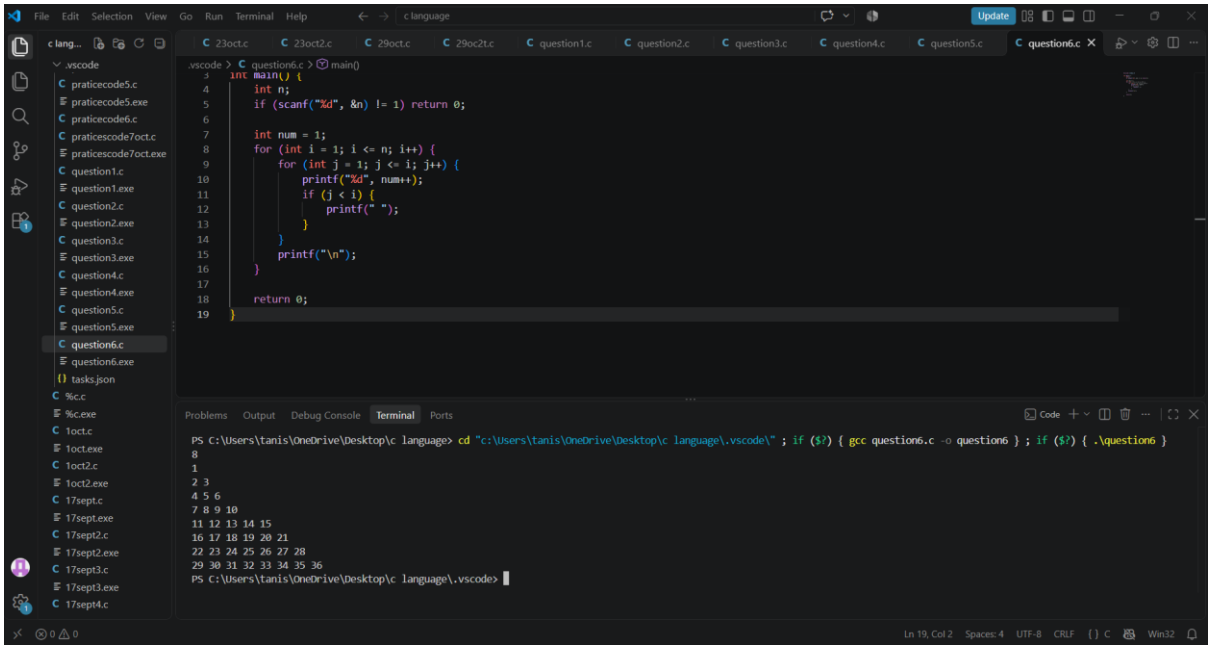
Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, with values separated by one space.

Sample Input	Sample Output
4	1 2 3 4 5 6 7 8 9 10

Hint: Maintain one counter that increases after each printed number.

ANSWER



Question 7 =

Question 7 — Print a Right-Aligned Star Triangle

Difficulty: Easy

Problem statement: Given n, print a right-aligned triangle. Row 1 has one asterisk and row n has n asterisks. Use leading spaces to align the right edges.

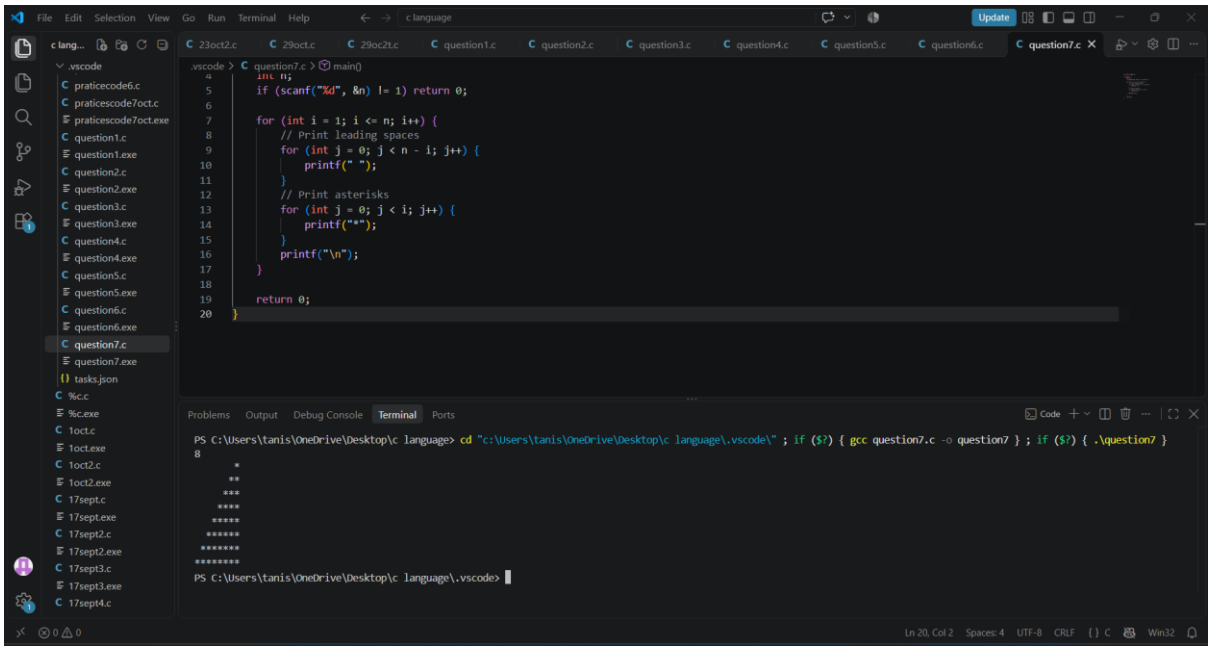
Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines. Leading spaces are required; do not add trailing spaces.

Sample Input	Sample Output
4	* ** *** ****

Hint: Row i has n - i leading spaces and i asterisks.

ANSWER



Question 8 =

Question 8 — Print a Centered Pyramid

Difficulty: Easy–Moderate

Problem statement: Given n, print a centered pyramid with n rows. Row i contains 2i - 1 asterisks, preceded by enough spaces to center it.

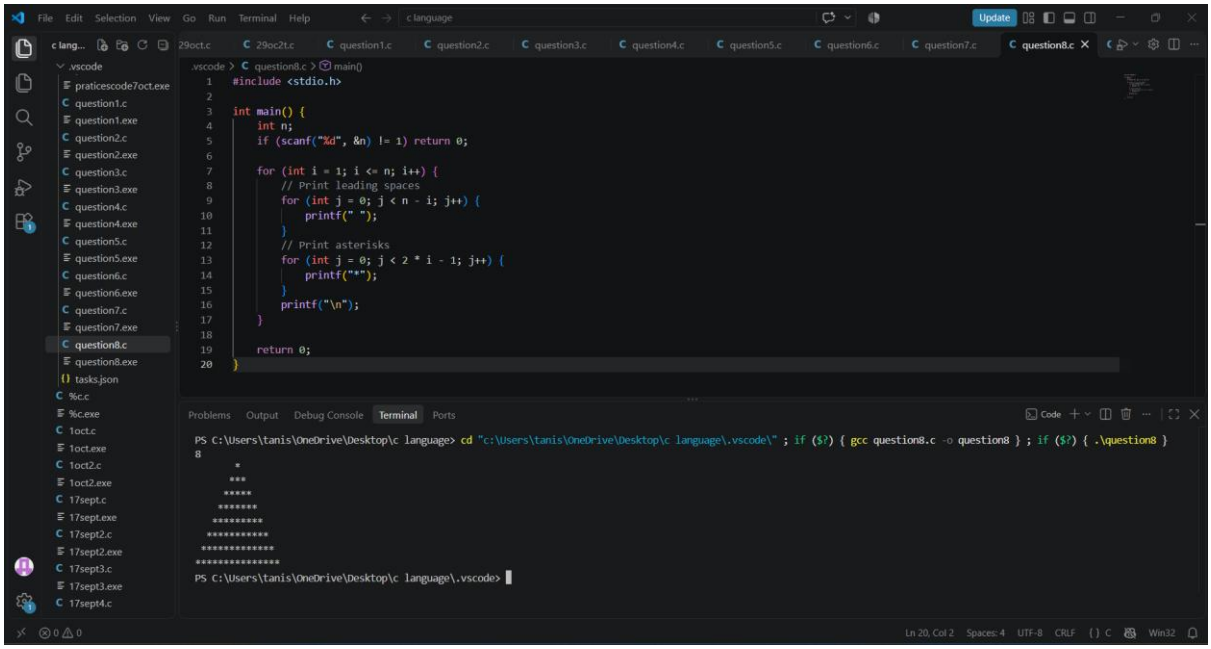
Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines of leading spaces followed by an odd number of asterisks.

Sample Input	Sample Output
4	* *** ***** *******

Hint: For row i, print n - i spaces and then 2*i - 1 asterisks.

ANSWER



Question 9 =

Question 9 — Print an Inverted Centered Pyramid

Difficulty: Easy–Moderate

Problem statement: Given n, print an inverted centered pyramid. The first row contains 2n - 1 asterisks; each next row has two fewer and one more leading space.

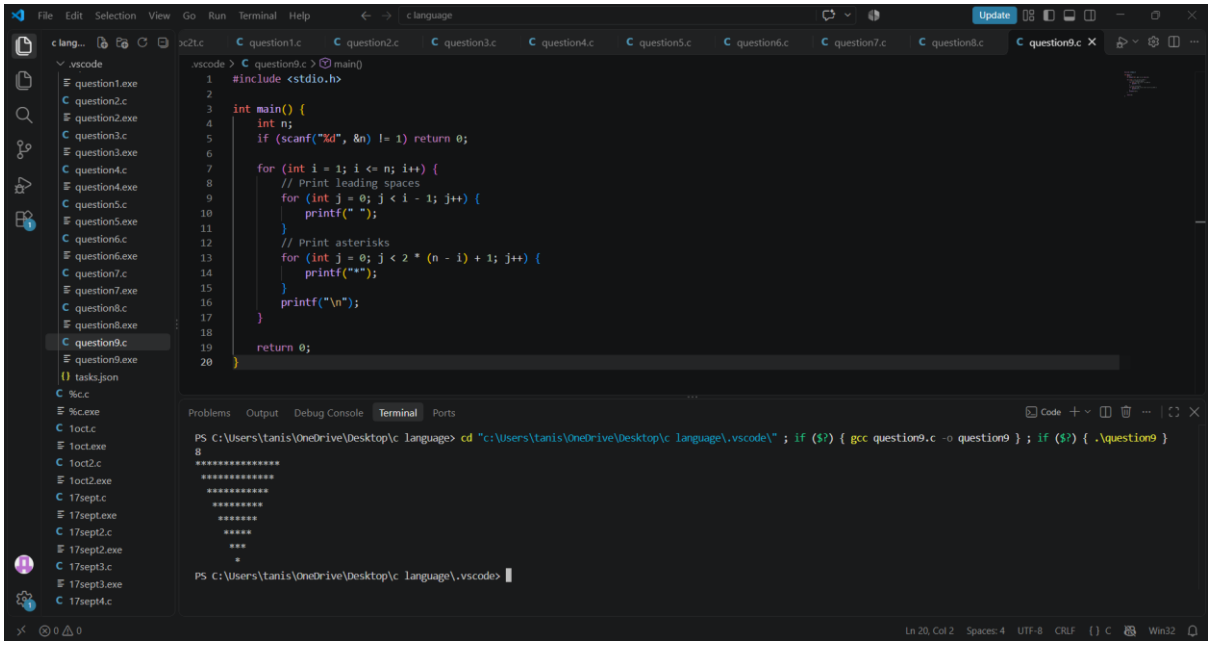
Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines, with the pyramid narrowing toward the bottom.

Sample Input	Sample Output
4	******* ***** *** *

Hint: For row i, print i - 1 spaces and 2*(n - i) + 1 asterisks.

ANSWER



Question 10 =

Question 10 — Print a Hollow Square

Difficulty: Moderate

Problem statement: Given n, print an n-by-n hollow square. Print an asterisk on the first row, last row, first column, or last column. Print a space inside.

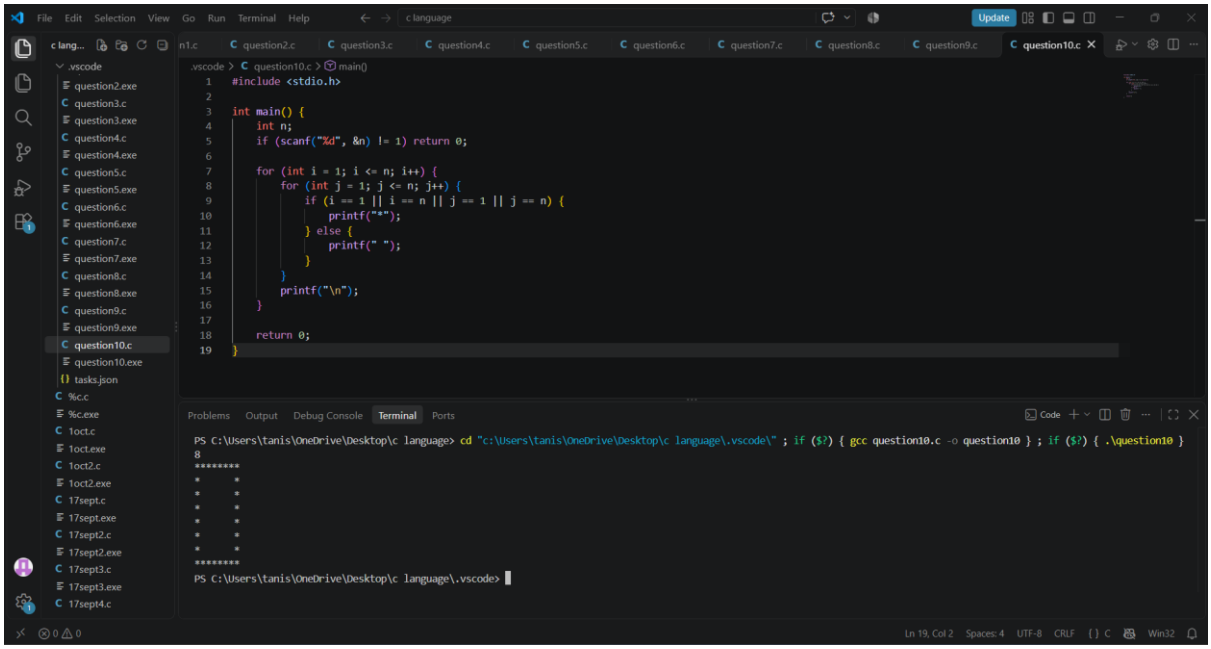
Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines of width n; interior positions are spaces.

Sample Input	Sample Output
5	***** * * * * * * *****

Hint: Print '*' when the row or column is on a boundary; otherwise print a space.

ANSWER



Question 11 =

Question 11 — Print a Hollow Right-Angled Triangle

Difficulty: Moderate

Problem statement: Given n, print the outline of a left-aligned right-angled triangle. Print asterisks in the first column, on the diagonal boundary, and across the final row. Print spaces inside.

Input format: One integer n (1 ≤ n ≤ 20).

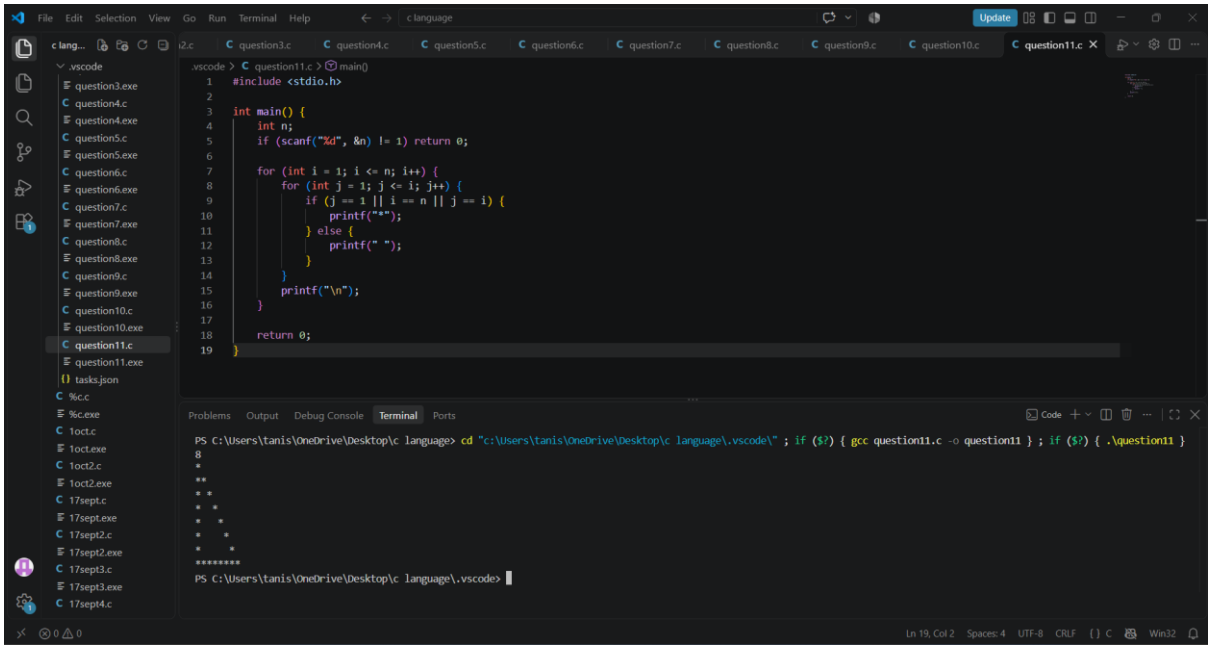
Output format: Print n lines. Preserve interior spaces; the final row contains n asterisks.

C Programming Practice Coding Platform Question Bank Set 1

Sample Input	Sample Output
5	* ** * * * * *****

Hint: A position is on the outline if it is in the first column, on the diagonal, or in the final row.

ANSWER



Question 12 =

Question 12 — Print a Multiplication Table Grid

Difficulty: Moderate

Problem statement: Given n, print an n-by-n multiplication grid. At row i and column j, print i multiplied by j. Separate values with one space.

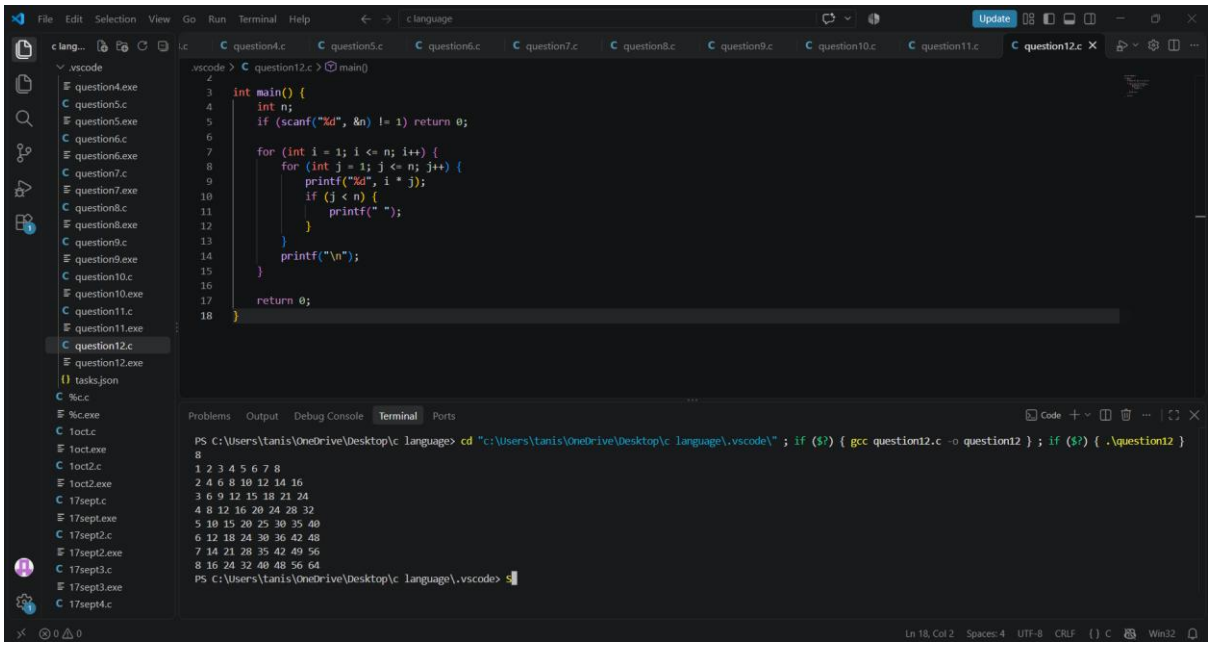
Input format: One integer n (1 ≤ n ≤ 12).

Output format: Print n lines, each containing n products separated by one space.

Sample Input	Sample Output
3	1 2 3 2 4 6 3 6 9

Hint: The value at each position is the row number multiplied by the column number.

ANSWER



Question 13 =

Question 13 — Print a Checkerboard

Difficulty: Moderate

Problem statement: Given n, print an n-by-n checkerboard using X and O. The top-left character must be X; adjacent horizontal and vertical cells alternate.

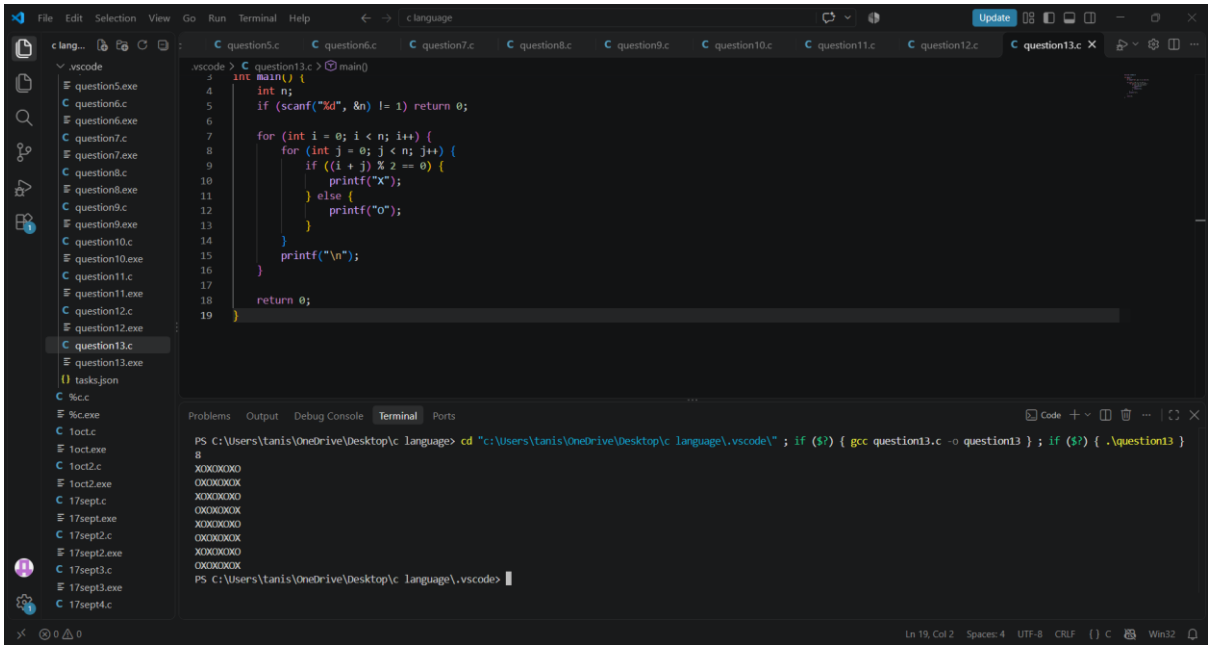
Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines of n characters, without spaces between characters.

Sample Input	Sample Output
4	XOXO OXOX XOXO OXOX

Hint: Use whether the sum of the row and column indices is even or odd.

ANSWER



Question 14 =

Question 14 — Print Floyd's Triangle

Difficulty: Easy–Moderate

Problem statement: Given n, print Floyd's triangle with n rows. Fill it with consecutive positive integers starting at 1, placing one number on row 1, two on row 2, and so on.

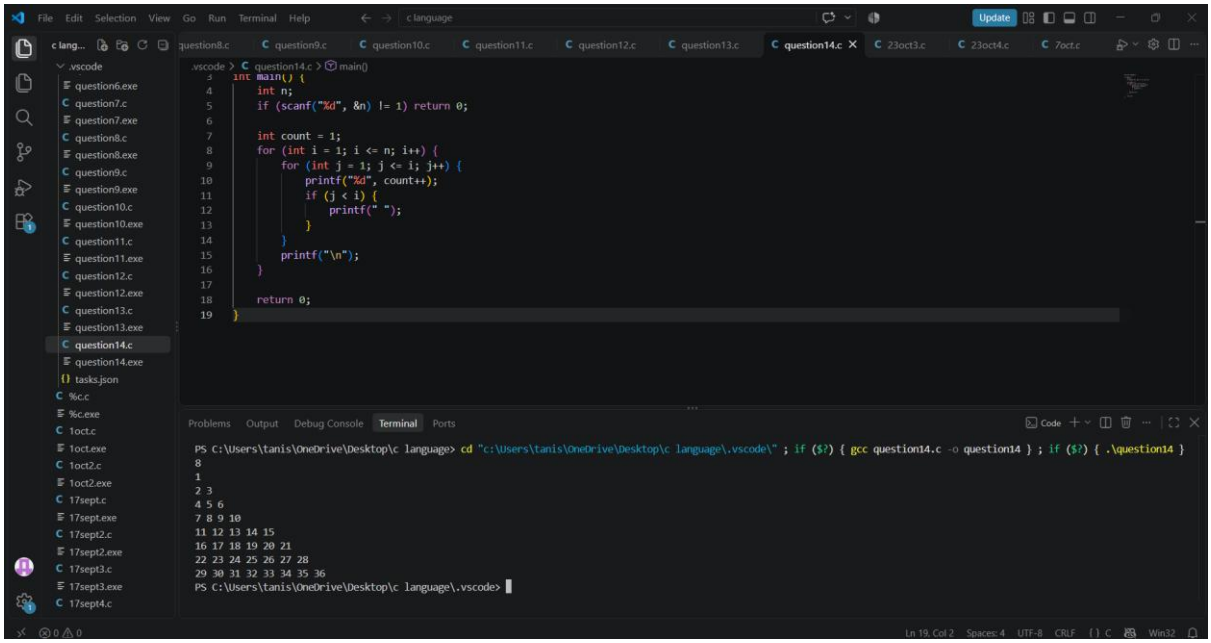
Input format: One integer n (1 ≤ n ≤ 20).

Output format: Print n lines with numbers separated by one space.

Sample Input	Sample Output
4	1 2 3 4 5 6 7 8 9 10

Hint: Use a counter that continues increasing across rows.

ANSWER



Question 15 =

Question 15 — Print a Butterfly Pattern

Difficulty: Moderate

Problem statement: Given an odd integer n, print a butterfly made of asterisks. The top half widens row by row and the lower half mirrors it. Use spaces between the left and right wings.

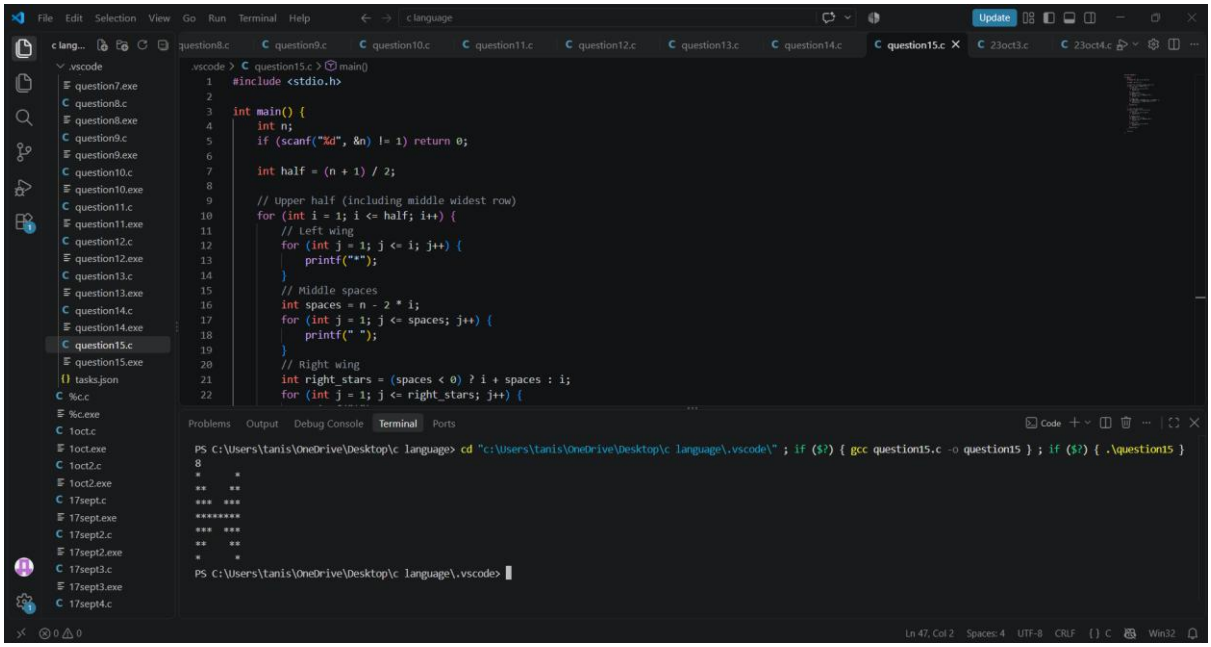
Input format: One odd integer n (3 ≤ n ≤ 19).

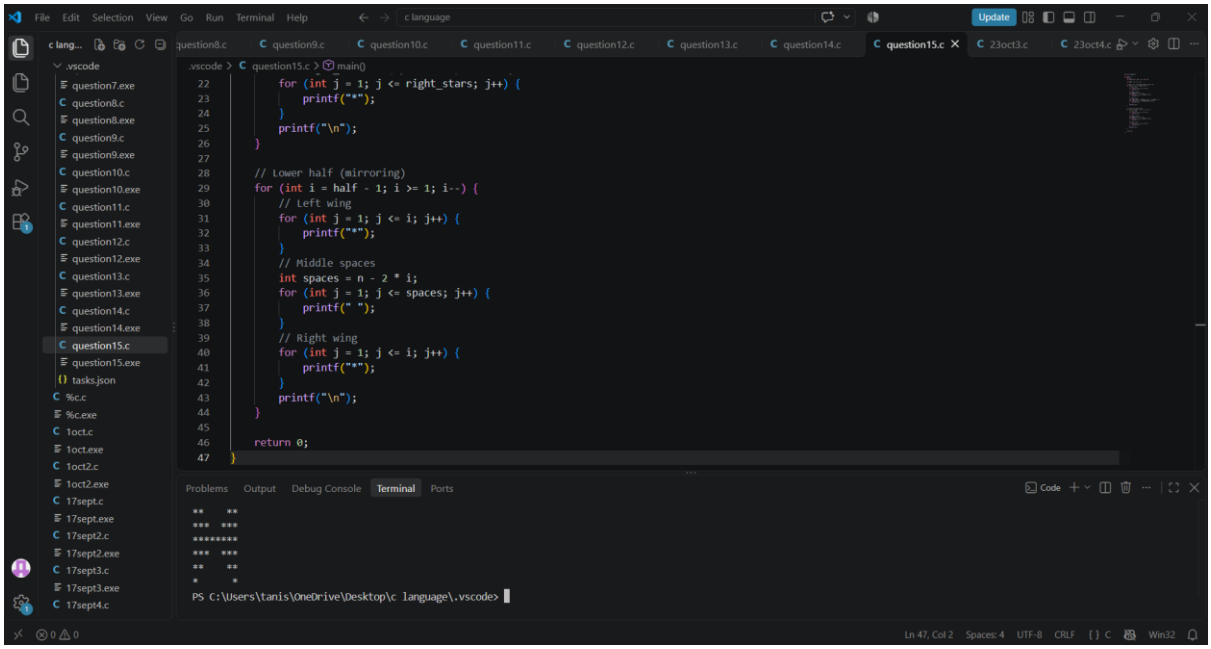
Output format: Print n lines. Each line contains left and right wings with a central gap.

Sample Input	Sample Output
5	* * ** ** *** *** ** ** * *

Hint: Build the upper half with increasing wing sizes, then mirror it.

ANSWER




File Edit Selection View Go Run Terminal Help
c language
Update
question8.c
question9.c
question10.c
question11.c
question12.c
question13.c
question14.c
question15.c
23oct3.c
23oct4.c
.vscode > question15.c > main()
for (int j = 1; j <= right_stars; j++) {
printf("*");
}
printf("\n");
}
// Lower half (mirroring)
for (int i = half - 1; i >= 1; i--) {
// Left wing
for (int j = 1; j <= i; j++) {
printf("*");
}
// Middle spaces
int spaces = n - 2 * i;
for (int j = 1; j <= spaces; j++) {
printf(" ");
}
// Right wing
for (int j = 1; j <= i; j++) {
printf("*");
}
printf("\n");
}
return 0;
}
Problems Output Debug Console Terminal Ports
Code
** **
*** ***
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*** ***
** **
* *
PS C:\Users\tanis\OneDrive\Desktop\c language\.vscode>
Ln 47, Col 2 Spaces: 4 UTF-8 CRLF { } C Win32

Question 16 =

Question 16 — Print a Diamond Pattern

Difficulty: Moderate

Problem statement: Given n, print a centered diamond. The upper half has n rows including the widest row; then print n - 1 rows in decreasing width.

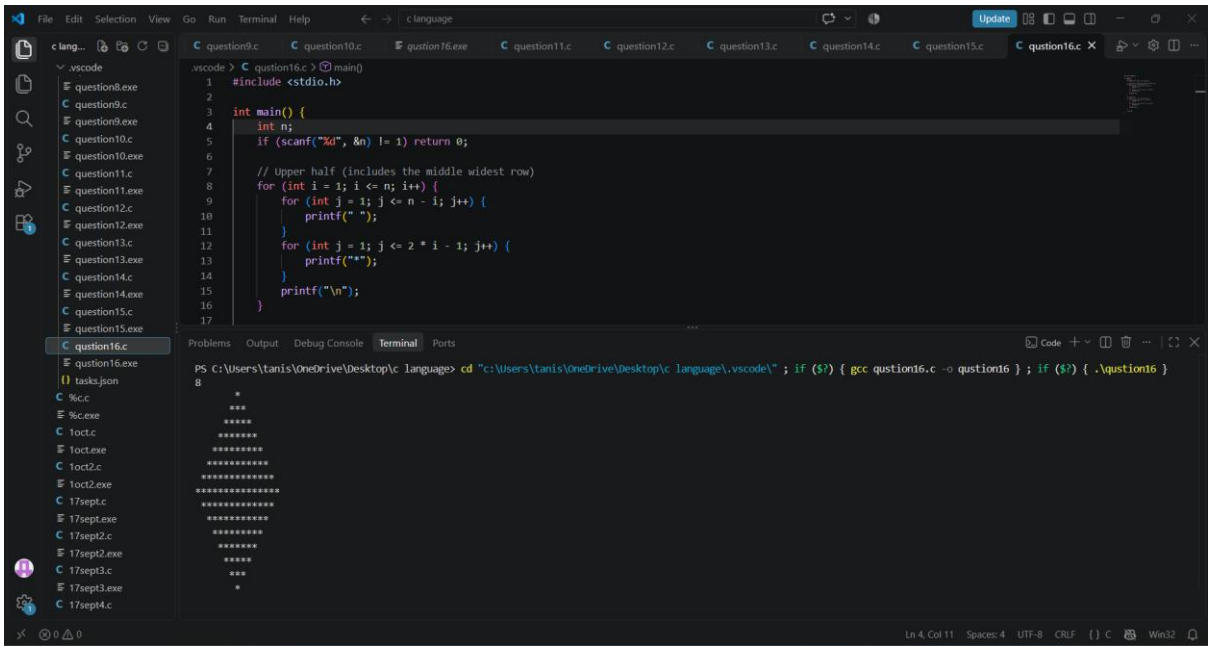
Input format: One integer n (1 ≤ n ≤ 10), representing the number of rows in the upper half.

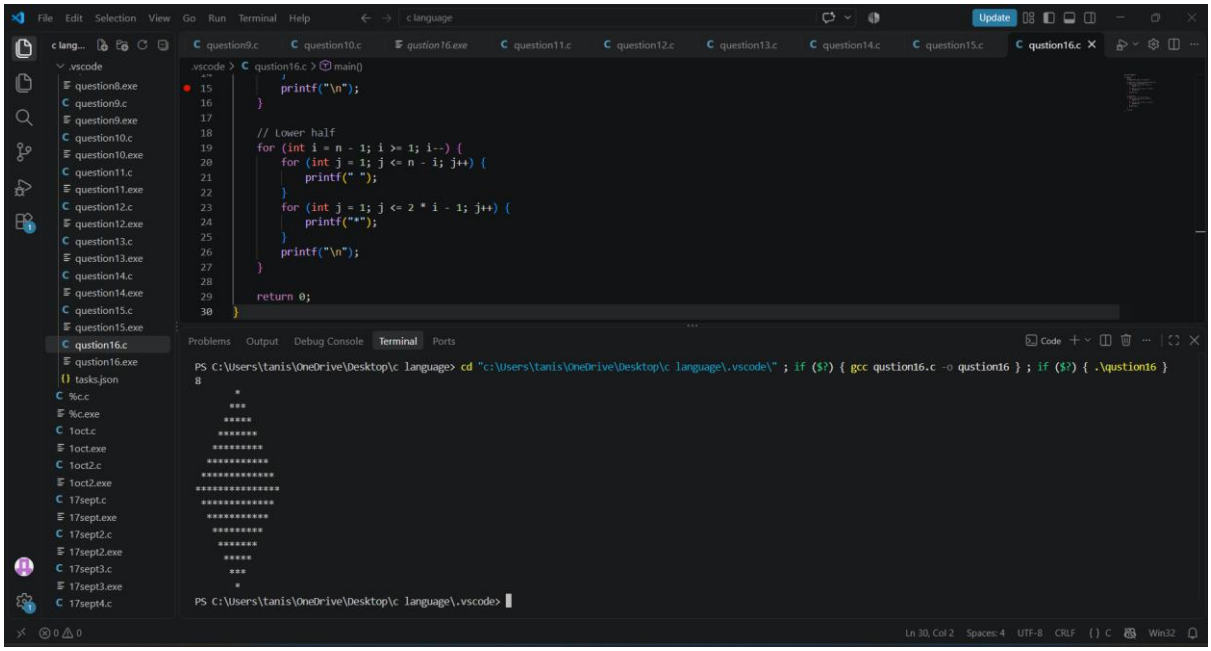
Output format: Print 2n - 1 centered lines.

Sample Input	Sample Output
3	* *** ***** *** *

Hint: Print an increasing pyramid, then a decreasing half without repeating the widest row.

ANSWER




File Edit Selection View Go Run Terminal Help
c language
Update
question9.c
question10.c
qustion16.exe
question11.c
question12.c
question13.c
question14.c
question15.c
qustion16.c
.vscode
question8.exe
question9.c
question9.exe
question10.c
question10.exe
question11.c
question11.exe
question12.c
question12.exe
question13.c
question13.exe
question14.c
question14.exe
question15.c
question15.exe
qustion16.c
qustion16.exe
tasks.json
%c.c
%c.exe
1oct.c
1oct.exe
1oct2.c
1oct2.exe
17sept.c
17sept.exe
17sept2.c
17sept2.exe
17sept3.c
17sept3.exe
17sept4.c
.vscode > qustion16.c > main()
printf("\n");
}
// Lower half
for (int i = n - 1; i >= 1; i--) {
for (int j = 1; j <= n - i; j++) {
printf(" ");
}
for (int j = 1; j <= 2 * i - 1; j++) {
printf("*");
}
printf("\n");
}
return 0;
}
Problems Output Debug Console Terminal Ports
Code
PS C:\Users\tanis\OneDrive\Desktop\c language> cd "c:\Users\tanis\OneDrive\Desktop\c language\.vscode\" ; if ($?) { gcc qustion16.c -o qustion16 } ; if ($?) { .\qustion16 }
8
*
***
*****
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*
PS C:\Users\tanis\OneDrive\Desktop\c language\.vscode>
Ln 30, Col 2
Spaces: 4
UTF-8
CRLF
{} C
Win32